

# URBAN GREEN SPACE MAPPING IN CHENNAI CITY

## Abstract

This research was carried out to map and analyze urban green space distribution in Chennai City using Geographic Information System (GIS) and remote sensing techniques. Vegetation cover in urban parks and surrounding built-up areas was assessed using the Normalized Difference Vegetation Index (NDVI) derived from Sentinel-2 multispectral satellite imagery. The Sentinel-2 data were processed and analyzed within a GIS framework to generate NDVI maps representing vegetation density and health across the study area. Urban green spaces, including public parks and vegetated open areas, were extracted and classified based on NDVI threshold values. Spatial analysis was performed to evaluate the distribution, extent, and variability of vegetation cover within different zones of the city. The results revealed significant spatial variation in vegetation density, with higher NDVI values corresponding to well-maintained urban parks and lower values associated with densely built-up areas. The accuracy of the NDVI-based vegetation mapping was validated through visual interpretation and comparison with high-resolution imagery. The study demonstrates the effectiveness of Sentinel-2 NDVI and GIS-based spatial analysis for urban green space mapping and provides valuable insights for urban planning, environmental management, and sustainable city development in Chennai.

**Keywords:** NDVI, Green Space, Land Use Land Cover, MSAVI, MNDWI, BI, NDBI, LST GIS

## Introduction

Rapid urbanization and population growth have significantly transformed land-use patterns in cities worldwide, leading to the continuous reduction and fragmentation of urban green spaces. Urban green spaces, including parks, gardens, street vegetation, and open green areas, play a vital role in maintaining ecological balance, improving air quality, regulating urban microclimates, and enhancing the physical and mental well-being of urban residents. However, uncontrolled urban expansion and infrastructure development exert increasing pressure on these green spaces, especially in rapidly growing metropolitan cities of developing countries. India is experiencing one of the fastest rates of urban growth globally, and metropolitan cities such as Chennai have undergone substantial land-use changes over recent decades. Chennai, one of the major urban centers in southern India, faces challenges such as loss of vegetation cover, increasing land surface temperature, air pollution, and reduced per capita green space availability. Monitoring and mapping urban green spaces are therefore essential for sustainable urban planning, climate resilience, and environmental management in the city.

Remote sensing and Geographic Information Systems (GIS) provide powerful tools for mapping, monitoring, and analyzing urban vegetation at multiple spatial and temporal scales. Among various vegetation indices, the Normalized Difference Vegetation Index (NDVI) has been widely used to assess vegetation density, health, and spatial distribution using multispectral satellite imagery. The availability of high-resolution and freely accessible Sentinel-2 satellite data has significantly enhanced the accuracy of urban green space mapping due to its fine spatial resolution and frequent revisit time. Several studies have employed GIS-based NDVI analysis to evaluate urban vegetation patterns, green space distribution, and their relationship with urban land-use dynamics. Some researchers have focused on temporal changes in vegetation cover, while others have assessed the role of green spaces in mitigating urban heat island effects and improving environmental quality. Despite these efforts, detailed spatial assessments of urban green spaces using Sentinel-2 NDVI at the city scale remain limited for Chennai, particularly with a focus on urban parks and organized green areas. Therefore, the present study aims to map and analyze urban green spaces in Chennai City using Sentinel-2 NDVI within a GIS framework. The study evaluates the spatial distribution and density of vegetation cover in urban parks and surrounding areas and provides a comprehensive assessment of green space patterns across the city. The results of this study are expected to support urban planners and policymakers in improving green infrastructure planning, promoting sustainable urban development, and enhancing environmental quality in Chennai.

## Study Area

Chennai City is located along the eastern coast of India in the state of Tamil Nadu, between 12°50' to 13°17' N latitude and 79°59' to 80°20' E longitude, bordering the Bay of Bengal to the east. Chennai serves as the capital city of Tamil Nadu and is one of the major metropolitan centers of southern India. The city covers an area of approximately 426 km<sup>2</sup> and is administratively divided into multiple zones and wards. The study area is bounded by Tiruvallur District to the north and west, Kanchipuram District to the south and southwest, and the Bay of Bengal to the east. Chennai is characterized by a predominantly flat coastal plain, with elevations ranging from sea level to about 60 m above mean sea level. The city is drained by three major rivers, namely the Cooum, Adyar, and Kosasthalaiyar, along with several canals and water bodies that influence urban vegetation distribution. The climate of Chennai is tropical, with a mean annual temperature of approximately 28–30°C. The city receives an average annual rainfall of about 1200–1400 mm, most of which occurs during the Northeast monsoon (October–December). High humidity and seasonal rainfall play a crucial role in sustaining urban vegetation and green spaces.

Soils in the study area are mainly sandy to clayey, particularly in coastal and riverine zones, with patches of alluvial and loamy soils in inland areas. Urban land use in Chennai consists of dense built-up areas interspersed with urban parks, institutional campuses, road-side plantations, wetlands, and remnant open green spaces. Major green spaces include public parks, recreational grounds, urban forests, and landscaped areas maintained by municipal authorities. Vegetation in Chennai is largely urban and semi-natural, consisting of tree species such as *Azadirachta indica* (Neem), *Ficus benghalensis*, *Ficus religiosa*, *Pongamia pinnata*, *Albizia lebbek*, and *Delonix regia*, along with shrubs and ornamental

plant species commonly found in parks and residential areas. These urban green spaces play an important role in improving air quality, regulating microclimate, and enhancing ecological sustainability in the rapidly urbanizing environment of Chennai.

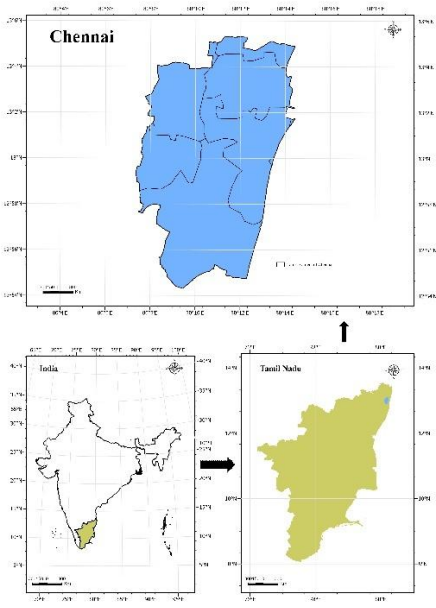


Fig. 1. Study area map

## Methodology

In this study, the methodology consisted of four main components:

1. Identification of urban green space indicators and data preparation, including the acquisition of Sentinel-2 multispectral satellite imagery and ancillary spatial data, and the preparation of thematic layers within a GIS environment;
2. Derivation of NDVI and classification of vegetation cover, where the Normalized Difference Vegetation Index (NDVI) was calculated from Sentinel-2 imagery and classified into different vegetation density classes to extract urban green spaces;
3. Spatial analysis and mapping of urban green spaces, involving the integration of NDVI results with land use/land cover data to generate urban green space maps and analyze their spatial distribution across Chennai City;
4. Validation and accuracy assessment of the urban green space maps, carried out through visual interpretation and comparison with high-resolution satellite imagery and existing land-use information.

*Identification of urban green space indicators and data preparation, including the acquisition of Sentinel-2 multispectral satellite imagery and ancillary spatial data, and the preparation of thematic layers within a GIS environment*

Urban green space indicators were identified based on the objectives of the study and relevant literature. Sentinel-2 Level-2A multispectral satellite imagery was acquired for Chennai City, along with ancillary spatial datasets such as administrative boundaries and land use information. All datasets were pre-processed, geometrically corrected, and organized within a GIS environment. Thematic layers required for the analysis were prepared and standardized for further processing.

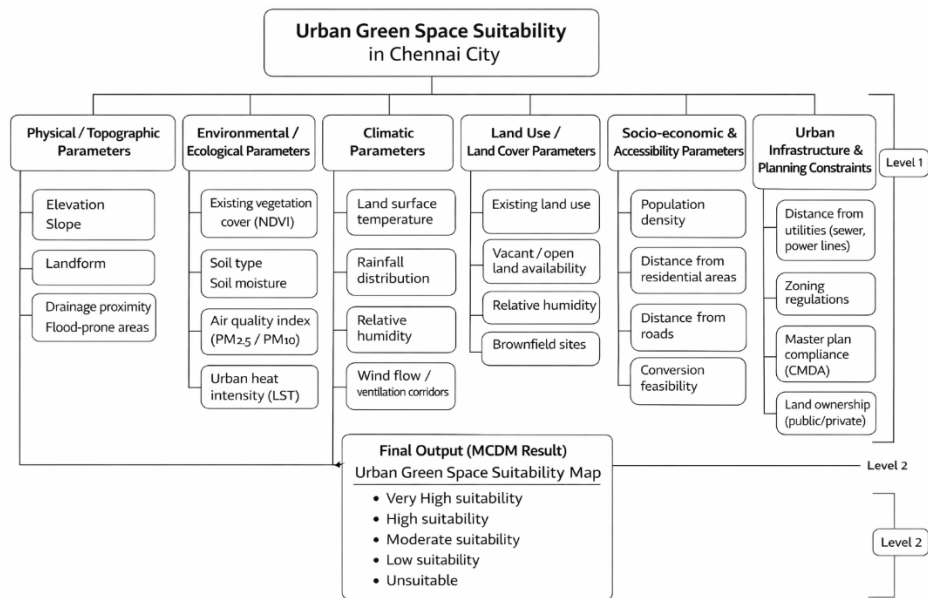


Fig. 2. Urban Green Space mapping criteria and sub-criteria in the study area.

*Derivation of NDVI and classification of vegetation cover, where the Normalized Difference Vegetation Index (NDVI) was calculated from Sentinel-2 imagery and classified into different vegetation density classes to extract urban green spaces*

The Normalized Difference Vegetation Index (NDVI) was derived from Sentinel-2 imagery using the red and near-infrared bands. The resulting NDVI values were classified into different vegetation density classes, representing dense, moderate, sparse, and non-vegetated areas. This classification enabled the extraction and identification of urban green spaces within the study area.

*Spatial analysis and mapping of urban green spaces, involving the integration of NDVI results with land use/land cover data to generate urban green space maps and analyze their spatial distribution across Chennai City*

The classified NDVI outputs were integrated with land use/land cover data within a GIS framework. Spatial analysis techniques were applied to generate urban green space maps and to examine their spatial distribution across Chennai City. This process facilitated the assessment of the extent and spatial pattern of urban green spaces.

*Validation and accuracy assessment of the urban green space maps, carried out through visual interpretation and comparison with high-resolution satellite imagery and existing land-use information*

Validation of the urban green space maps was carried out through visual interpretation and comparison with high-resolution satellite imagery and existing land use information. This accuracy assessment ensured the reliability of the results and confirmed the consistency of the extracted green spaces with actual ground conditions.

## Results

The spatial analysis of urban green spaces in Chennai City was carried out using Sentinel-2 multispectral imagery and multiple spectral indices, including NDVI, SAVI, MSAVI, NDBI, MNDWI, Bareness Index, and Land Surface Temperature (LST). The results reveal significant spatial variability in vegetation cover, built-up intensity, surface temperature, and land use patterns across the study area. The Normalized Difference Vegetation Index (NDVI) map indicates that vegetation cover in Chennai City is generally low to moderate, with NDVI values ranging approximately from  $-0.33$  to  $0.69$ . Higher NDVI values, representing dense vegetation, are mainly concentrated in urban parks, institutional campuses, river corridors, wetlands, and peripheral zones, particularly in the southern and south-western parts of the city. In contrast, the central and northern parts of Chennai exhibit low NDVI values, reflecting dense built-up areas and limited green cover.

The Soil Adjusted Vegetation Index (SAVI) and Modified Soil Adjusted Vegetation Index (MSAVI) maps further confirm the sparse and fragmented nature of vegetation within the urban core. These indices effectively reduce soil background effects and highlight isolated vegetation patches within heavily urbanized zones. Higher SAVI and MSAVI values are observed in peri-urban agricultural lands, open spaces, and green institutional areas, whereas low values dominate compact residential and industrial regions. The Land Use Land Cover (LULC) classification shows that built-up areas form the dominant land cover class, followed by scattered patches of vegetation, wetlands, and water bodies. Urban green spaces are unevenly distributed and largely confined to specific land use categories such as parks, plantations, wetlands, and cultivated lands. Rapid urban expansion has resulted in the fragmentation and reduction of continuous green spaces, especially in the city's core areas.

The Normalized Difference Built-up Index (NDBI) map highlights extensive built-up surfaces across Chennai City, with higher values concentrated in the central business districts, industrial zones, and major transportation corridors. Areas with high NDBI values correspond closely with regions showing low NDVI and SAVI values, indicating an inverse relationship between built-up density and vegetation cover. The Bareness Index map indicates widespread exposed and impervious surfaces across the city, particularly in newly developed and construction-prone areas. These regions often overlap with high NDBI and LST zones, reflecting land cover conversion and reduced ecological quality.

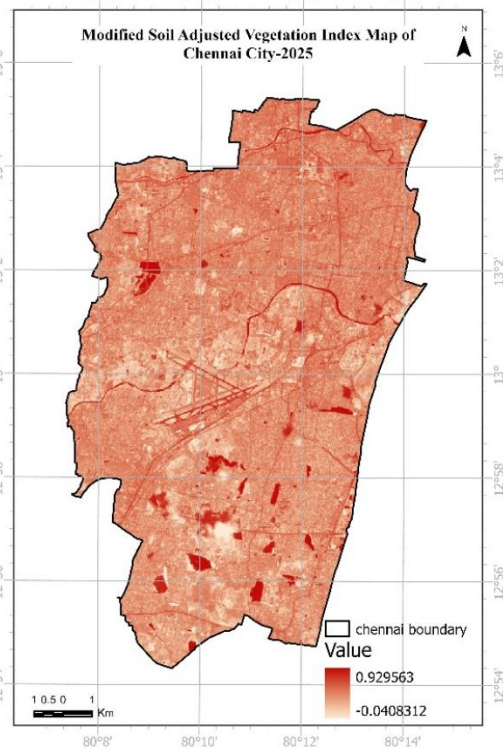
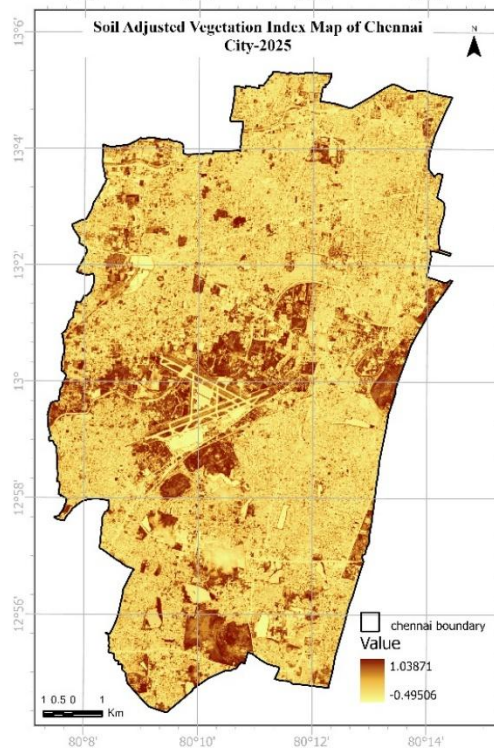
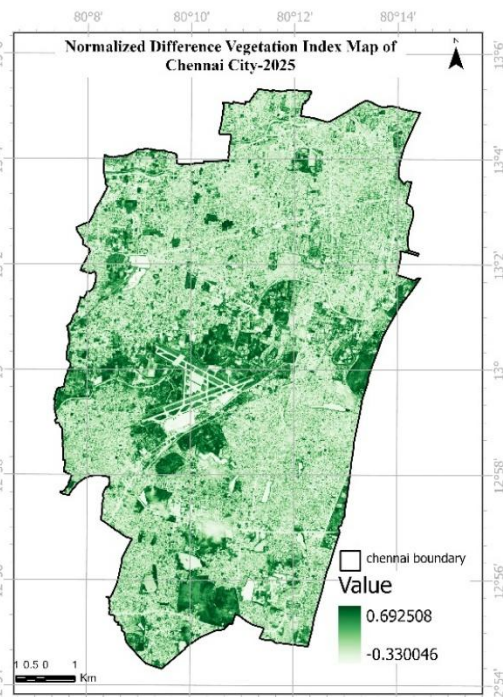
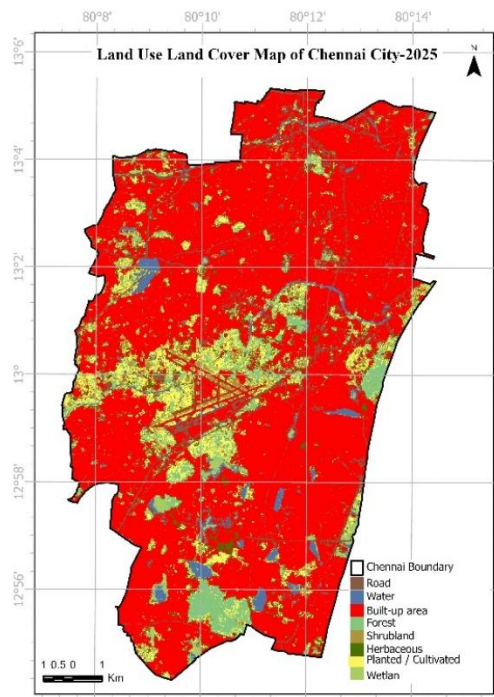
The Land Surface Temperature (LST) map shows elevated surface temperatures across densely built-up areas, while relatively lower temperatures are observed in regions with higher vegetation density and water bodies. This spatial pattern clearly demonstrates the cooling effect of urban green spaces and their role in mitigating the urban heat island effect. The Modified Normalized Difference Water Index (MNDWI) map effectively delineates water bodies, wetlands, and drainage networks within Chennai City. Areas with higher MNDWI values are associated with lower LST and moderate vegetation presence, emphasizing the importance of blue-green infrastructure in urban climate regulation. Overall, the integration of vegetation indices, built-up indicators, and thermal analysis reveals that urban green spaces in Chennai City are spatially fragmented, unevenly distributed, and under pressure from rapid urbanization.

## Discussion

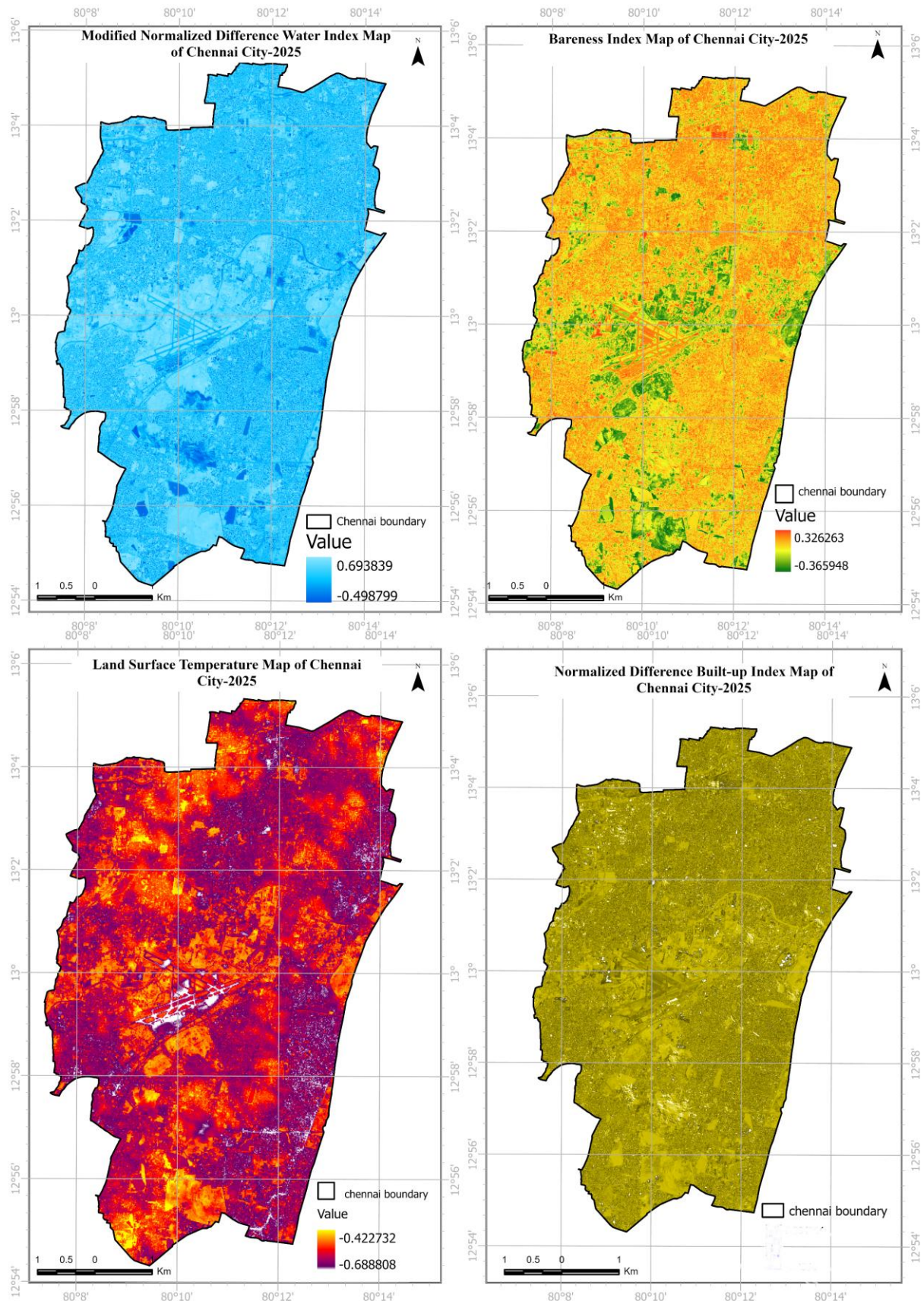
The land use and land cover statistics reveal that **built-up areas dominate Chennai City**, covering approximately **124.54 km<sup>2</sup>**, which clearly explains the widespread occurrence of high NDBI values, elevated bareness, and increased land surface temperatures observed in the spatial analysis. This extensive built-up dominance corresponds closely with low NDVI, SAVI, and MSAVI values across central and northern parts of the city, indicating limited and fragmented vegetation cover. In contrast, vegetated land use classes—including **planted/cultivated land (12.38 km<sup>2</sup>)**, **forest (6.99 km<sup>2</sup>)**, **shrubland (6.32 km<sup>2</sup>)**, and **herbaceous cover (5.49 km<sup>2</sup>)**—together account for a comparatively smaller proportion of the urban landscape and are mainly concentrated in peri-urban areas, institutional campuses, and along water bodies. These zones consistently exhibit higher vegetation index values and relatively lower surface temperatures, highlighting the cooling and ecological benefits of urban green spaces. Water bodies (**3.93 km<sup>2</sup>**) and wetlands (**5.30 km<sup>2</sup>**) play a critical role in urban environmental regulation, as reflected by higher MNDWI values and localized reductions in land surface temperature. However, their limited spatial extent underscores growing pressure from urban

expansion. Road networks, occupying **12.07 km<sup>2</sup>**, further contribute to surface sealing and heat accumulation, reinforcing the inverse relationship between impervious surfaces and vegetation cover. Overall, the combined interpretation of LULC statistics and spectral index maps demonstrates that rapid urbanization has significantly reduced continuous green spaces in Chennai City, emphasizing the urgent need for strategic green infrastructure planning and conservation measures.

| Sl.No | Land Type            | Area(Square Kilometer) |
|-------|----------------------|------------------------|
| 1     | Built-up Area        | 124.536                |
| 2     | Forest               | 6.99519                |
| 3     | Herbaceous           | 5.49452                |
| 4     | Planted / Cultivated | 12.3798                |
| 5     | Road                 | 12.0703                |
| 6     | Shrubland            | 6.31819                |
| 7     | Water                | 3.9323                 |
| 8     | Wetlands             | 5.29699                |







## Conclusions

This study successfully demonstrated the application of Sentinel-2 multispectral satellite data and GIS-based spatial analysis for urban green space mapping in Chennai City. The results indicate that although green spaces exist across the city, their distribution is highly uneven, with dense vegetation largely restricted to peripheral areas, water bodies, institutional lands, and protected zones. Central and northern parts of the city are dominated by built-up surfaces, high bareness, and elevated land surface temperatures, reflecting intense urbanization and limited ecological cover. The inverse relationship observed between vegetation indices (NDVI, SAVI, MSAVI) and built-up indicators (NDBI and Bareness Index), along with the spatial correspondence between vegetation cover and lower surface temperatures, highlights the critical role of urban green spaces in mitigating urban heat stress and improving environmental quality. The LULC analysis further emphasizes the fragmentation of green spaces due to expanding built-up land use.

The findings underline the need for strategic urban green space planning, particularly in high-density and high-temperature zones of Chennai City. Integrating green infrastructure into urban development plans can enhance thermal comfort, environmental sustainability, and overall urban liveability. The methodology adopted in this study provides a reliable and transferable framework for urban green space assessment and can support urban planners and decision-makers in developing sustainable city planning strategies. Future studies may incorporate higher-resolution imagery, ground-based validation, and socio-economic indicators to further refine urban green space assessments and evaluate their accessibility and ecosystem services.

### Policy Implications for Urban Planning in Chennai

- 1. Prioritize Protection of Existing Green Assets**  
Areas exhibiting high vegetation indices and lower surface temperatures should be designated as protected urban green zones, with strict controls on land use conversion and encroachment.
- 2. Target Green Space Expansion in Heat- and Density-Stressed Zones**  
High built-up and high land surface temperature areas should be prioritized for green space interventions, including neighborhood parks, roadside plantations, and vertical greening solutions.
- 3. Integrate Green Infrastructure into Statutory Planning Frameworks**  
Urban green spaces should be mandated within master plans and development regulations, requiring minimum green cover standards for new and redeveloped urban projects.
- 4. Strengthen Blue–Green Network Conservation**  
Water bodies, wetlands, and associated green buffers should be conserved and interconnected to enhance flood regulation, urban cooling, and ecological resilience.
- 5. Institutionalize GIS-Based Monitoring for Adaptive Planning**  
Regular remote sensing–based monitoring of vegetation cover and land surface temperature should be adopted to support evidence-based planning and periodic policy evaluation.

### References

1. ALLGOWER B., CARLSON J.D., WAGTENDONK J.W.V., IntUoGXFtion to fiUH GangHU Uating anG UHmotH VHnVing- wiOO UHmotH VHnVing HnhanFH wiOG OanG fiUH GangHU Uating? In: WiOG OanG fiUH GangHU HVtimation anG maSSing, thH UoOH of UHmotH VHnVing Gata. (EGV ChXviHFo E.), WoUOG SFiHntifiF SXbOiVhing: NHw JHUVHy, **2003**.

2. BONAZOUNTAS M., KALLIDROMITOU D., KASSOMENOS P.A., PASSAS N., jiUH UiVN anaOyViV. HXm. EFoO. RiVN AVVHV. **1**, 617, **2005**.

3. CHUVIECO E., CONGALTON R.G., ASSOiFation of UHmotHVHnVing anG gHogUaShiF infoUation VyVtHmV to foUHVt fiUH hazaUG maSSing. RHmotH SHnV. EnviUon. **29**, 147, **1989**.

4. DONG X.U., LI-MIN D., GUO-)AN SH., LEI T., HUI W.,)oUHVt fiUH UiVN zonH maSSing fUom VatHOOitH imagHV anG GIS foU BaihH )oUHVtUy BXUHAX, JiOin, China. JoXUnaO of )oUHVtUy RHVHaUFh. **16** (3), 169, **2005**

5. SALAMATI H., MOSTA)ALOU H., MASTOORI A., HONARDOOST ), EvaOXation anG SUoviVion of foUHVt fiUH UiVN maS XVing GIS in GoOHVtan foUHVtV. PUoFHHGing of thH fiUVt intHUnationaO FonfHUHnFHV on fiUH in natXUaO UHVoxUFHV, IUan. **10**, **2011**.

6. MAHDAVI A., )ALLAHSHAMSI R., NAZARI R., HEIDARI M., PUoviVion of fiUH UiVN maS in foUHVt anG SaVtXUH OanGV in IOam Fity XVing AHP. PUoFHHGing of thH fiUVt intHUnationaO FonfHUHnFHV on fiUH in natXUaO UHVoxUFHV, IUan. **13**, **2011**.

7. MazanGaUan NatXUaO RHVoXUFHV AGminiVtUation (MNRA),)oUHVtUy SOan of DiVtUiFt thUHH of NHNa-ZaOHmUoXG foUHVtV, Ilan"V )oUHVtV, RangHOanG anG WatHUVhHG oUganization PXbOiVhing: MazanGaUan, **2010**.

8. ESKANDARI S., OLADI GHADIKOLAEI J., JALILVAND H., SARADJIAN M.R., )oUHVt jiUH RiVN MoGHOing anGPUHGifTion in DiVtUiFt ThUHH of NHNa-ZaOHmUoXG )oUHVt XVing GHogUaShiFaO InfoUation SyVtHm. IUanian JoXUnaO of )oUHVt anG PoSOaU RHVHaUFh. **21** (2), 203, **2013**.